

Lab Experiment 8: Conduct Z test for the Data Given using Python Objective.

Aim:

To test whether the average weight of a species of birds differs from 150 grams.

Procedure:

1. Null Hypothesis (H_0): The average weight of the birds is 150 grams.
2. Alternative Hypothesis (H_1): The average weight of the birds is not 150 grams.
3. Sample: Measure the weights of 30 birds randomly selected from the population.
4. Z-Test: Conduct a Z-test to compare the sample mean to 150 grams.
5. Decision Rule: Use a significance level of $\alpha = 0.05$.

Code:

```
import numpy as np
import scipy.stats as stats

sample_data = np.array([145, 148, 150, 147, 149, 151, 146, 152, 150, 148,
                        149, 147, 153, 150, 151, 146, 149, 152, 148, 150,
                        147, 151, 149, 150, 146, 153, 148, 151, 150, 149])

population_mean = 150

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data, ddof=1)

n = len(sample_data)

z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))

p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic)))

print(f"Sample Mean: {sample_mean:.2f}")

print(f"Z-Statistic: {z_statistic:.4f}")

print(f"P-Value: {p_value:.4f}")

alpha = 0.05

if p_value < alpha:

    print("Reject the null hypothesis: The average weight is significantly different from 150 grams.")
```

else:

```
print("Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.")
```

Output:

Sample Mean: 149.17

Z-Statistic: -2.1544

P-Value: 0.0312

Reject the null hypothesis: The average weight is significantly different from 150 grams.

Result:

There is a significant difference between the average bird weight and 150 grams at the 5% significance level.